INTRODUCTION TO COMPUTER AND IT'S APPLICATION

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INTRODUCTION TO COMPUTERS

- A computer is an electronic device that processes data and performs tasks according to a set of instructions called programs. It is composed of hardware (physical components) and software (programs and applications).
- It is a versatile tool that has become an integral part of our daily lives, helping us solve problems, communicate, and automate tasks.

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CHARACTERISTICS OF COMPUTERS:

- **Speed:** Computers can perform millions of calculations per second, measured in Hertz (Hz).
- **Accuracy:** They provide precise results, provided the input and instructions are correct.
- **Storage:** Computers can store vast amounts of data in secondary storage devices like hard disks and SSDs.
- **Automation:** Once programmed, computers can perform tasks automatically without human intervention.
- **Versatility:** They can perform a wide range of tasks, from simple calculations to complex simulations.

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TYPES OF COMPUTERS

1. **Supercomputers**:
    1. Used for complex calculations and simulations (e.g., weather forecasting, nuclear research).
    2. Examples: IBM Summit, Fugaku.
2. **Mainframe Computers**:
    1. Handle large-scale data processing for organizations (e.g., banks, airlines, governments).
    2. Known for their reliability and ability to process millions of transactions daily.
3. **Personal Computers (PCs)**:
    1. Designed for individual use (e.g., desktops, laptops).
    2. Used for work, education, entertainment, and more.

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4. **Mobile Devices**:
    1. Smartphones, tablets, and wearables (e.g., smartwatches).
    2. Portable and convenient for on-the-go use.
5. **Embedded Systems**:
    1. Computers integrated into other devices (e.g., cars, home appliances, medical equipment).
    2. Perform specific functions within larger systems.

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COMPONENTS OF A COMPUTER

Hardware:

- **CPU (Central Processing Unit)**: The "brain" of the computer that performs calculations and executes instructions.
- **Memory (RAM)**: Temporary storage for active programs and data.
- **Storage**: Devices like hard drives and SSDs that store data permanently.
- **Input Devices**: Tools like keyboards, mice, and scanners that allow users to interact with the computer.
- **Output Devices**: Monitors, printers, and speakers that display or deliver results.

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Software:

•**Operating System (OS)**: Manages hardware and software resources (e.g., Windows, macOS, Linux).

•**Applications**: Programs designed for specific tasks (e.g., Microsoft Word, Google Chrome, Photoshop)

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APPLICATIONS OF COMPUTERS

Education:

- Online learning platforms (e.g., Coursera, Khan Academy).
- Research and access to digital libraries.
- Virtual classrooms and video lectures.

Business:

- Accounting and payroll management.
- Inventory tracking and supply chain management.
- Customer relationship management (CRM) systems.

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Healthcare:

•Electronic health records (EHRs) for patient data management.

•Medical imaging and diagnostics (e.g., MRI, X-rays).

•Telemedicine for remote consultations.

Entertainment:

•Gaming (e.g., PC games, consoles).

•Streaming services (e.g., Netflix, Spotify).

•Digital art and animation.

Communication:

•Email, social media, and instant messaging.

•Video conferencing tools (e.g., Zoom, Microsoft Teams).

Science and Research:

•Data analysis and simulations.

•Modeling complex systems (e.g., climate change, space exploration).

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